

AP CALCULUS AB - UNIT 3

Guided Notes 3.3: Inverse Function and Inverse Trigonometric Derivatives



Name: _____ Class: _____ Date: _____

Learning Objectives and Essential Question

Essential question: How can we use the derivative-of-an-inverse formula from equations, tables, and graphs?

- Use the derivative-of-an-inverse formula from equations, tables, and graphs.
- Differentiate inverse trigonometric functions and compositions containing them.
- Write tangent lines to inverse functions and interpret reciprocal slopes.
- Combine inverse derivatives with the Chain, Product, and Quotient Rules.

Key Knowledge, Vocabulary, and Notation

Inverse derivative

If $f(a) = b$ and $f'(a) \neq 0$, then $(f^{-1})'(b) = 1/f'(a) = 1/f'(f^{-1}(b))$.

Reciprocal slope

A function and its inverse have reciprocal tangent slopes at reflected points.

Arcsine

$$\frac{d}{dx} \arcsin u = \frac{u'}{\sqrt{1-u^2}}.$$

Arccosine

$$\frac{d}{dx} \arccos u = -\frac{u'}{\sqrt{1-u^2}}.$$

Arctangent

$$\frac{d}{dx} \arctan u = \frac{u'}{1+u^2}.$$

Domain awareness

Square-root denominators require valid inputs and may vanish at domain endpoints.

Concept Development and Strategy

1. To find $(f^{-1})'(b)$, first identify the preimage a satisfying $f(a) = b$.
2. Evaluate f' at the preimage, not at the output b , then take the reciprocal.
3. For inverse trigonometric compositions, multiply the standard derivative by the derivative of the inside.
4. In a table, locate the output in the $f(x)$ row before reading the corresponding derivative entry.
5. Use radians for trigonometric derivative formulas.

Shared Representation / Data

x	-2	-1	0	1	3
$f(x)$	-5	-2	1	4	10
$f'(x)$	4	3	2	5	7

Worked Example: Inverse derivative from a formula

Let $f(x) = x^3 + x$. Find $(f^{-1})'(2)$.

Answer and Reasoning

Since $f(1) = 2$ and $f'(1) = 4$, $(f^{-1})'(2) = 1/4$.

Worked Example: Arcsine composition

Differentiate $y = \arcsin(3x)$.

Answer and Reasoning

$$y' = \frac{3}{\sqrt{1-9x^2}}.$$

Worked Example: Arctangent composition

Differentiate $y = \arctan(x^2)$.

Answer and Reasoning

$$y' = \frac{2x}{1+x^4}.$$

Guided Practice and Discussion

Directions

Show the mathematical setup, preserve exact values unless a decimal is requested, and justify existence or interpretation statements with the appropriate definition or theorem.

1. If $f(2) = 7$ and $f'(2) = 5$, find $(f^{-1})'(7)$.

2. Differentiate $y = \arcsin x$.

3. Let $f(x) = x^3 + 2x$. Find $(f^{-1})'(3)$.

4. Write the tangent line to $y = \arcsin x$ at $x = 1/2$.

5. Differentiate $y = \arcsin\left(\frac{x}{1+x}\right)$.

6. Find a so the tangent to $y = \arctan(ax)$ at $x = 0$ has slope 6.

7. Prove the inverse-derivative formula from $f(f^{-1}(x)) = x$.

AP Reasoning Check

Multiple-Choice Check

Choose the best answer. Explain why at least one distractor is incorrect during class discussion.

1. If $f(4) = 9$ and $f'(4) = -3$, then $(f^{-1})'(9)$ is

(A) -3

(B) $-1/3$

(C) $1/3$

(D) 3

2. The derivative of $\arctan(2x)$ is

(A) $2/(1 + 4x^2)$

(B) $1/(1 + 2x^2)$

(C) $2/\sqrt{1 - 4x^2}$

(D) $-2/(1 + 4x^2)$

3. Let $f(x) = x + e^x$. If $f(a) = 4$, then $(f^{-1})'(4)$ is closest to

(A) 0.119

(B) 0.255

(C) 0.500

(D) 0.731

Common Errors and AP Precision

- Mark every layer or every occurrence of y before differentiating; omitted inner derivatives are the most common source of error.
- For inverse derivatives, find the preimage first. For implicit relations, verify the point lies on the original curve.
- A complete AP justification names the relevant definition, condition, or theorem rather than reporting only a numerical result.

Exit Ticket

Differentiate and simplify $y = \arctan\left(\frac{1-x}{1+x}\right)$.

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